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loops

→ loop is a programming structure that repeats block of code until a condition becomes false or until all items are processed.

for loop

→ for loop is used when we know how many times we want to repeat

syntax - for variable in sequence
statement

```
a = [1, 2, 3, 4, 5]
```

```
for i in a:  
    print(i+1)
```

```
// 2 3 4 5 6
```

```
tup = (1, 7, 9, 6, 4, 2)
```

```
for i in tup:
```

```
    if i % 2 == 0:
```

```
        print(f"{i} is a even number")
```

while loop

→ a while loop runs as long as condition is true
syntax - while condition:
statement

```
count = 0
```

```
while count < 5: // 0 1 2 3 4
```

```
    print(count)
```

```
    count += 1
```

```
for i in range(3)
```

```
    for j in range(2)
```

```
        print(i, j)
```

```
// 0 0  
   0 1  
   1 0  
   1 1  
   2 0  
   2 1
```

Break

It stops the loop once it is true

```
for n in range(1, 10)
```

```
    if  $n-1 \cdot 3 == 0$ 
```

```
        print("Round", n)
```

```
        break
```

// found 3

Continue

It skips the condition

```
for i in range(5)
```

```
    if  $i == 2$ :
```

```
        continue
```

```
    print(i)
```